

Structural Graph Coverage Criteria

Meenakshi D'Souza

International Institute of Information Technology
Bangalore.

Graph coverage criteria: Overview

- We look at graphs as structures and look at the following two coverage criteria over graphs.
  - Structural coverage criteria.
  - Data flow coverage criteria.
- Later, we consider software artifacts (code, design elements and requirements) modelled as graphs and see how these coverage criteria apply to them.
- Focus of this lecture: Structural coverage criteria.

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Structural coverage criteria over graphs

- Node/vertex coverage.
- Edge coverage.
- Edge pair coverage.
- Path coverage
  - Complete path coverage.
  - Prime path coverage.
  - Complete round trip coverage.
  - Simple round trip coverage.

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Node coverage

- Node coverage requires that the test cases visit each node in the graph once.
- Node Coverage: Test set T satisfies node coverage on graph G iff for every syntactically reachable node $n \in G$, there is some path p in $path(T)$ such that p visits n.
- Simpler definition: TR contains each reachable node in G.

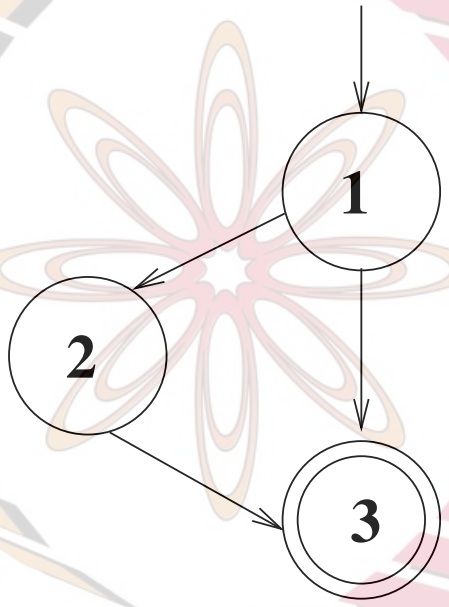
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Edge coverage

- Edge Coverage: *TR* contains each reachable path of length up to 1, inclusive, in G.
- Edge coverage is slightly stronger than node coverage.
- Why "length up to 1" ?
  It allows edge coverage for graphs with one node and no edges.
- Allowing length up to 1 allows edge coverage to subsume node coverage.

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Node and edge coverage: Example

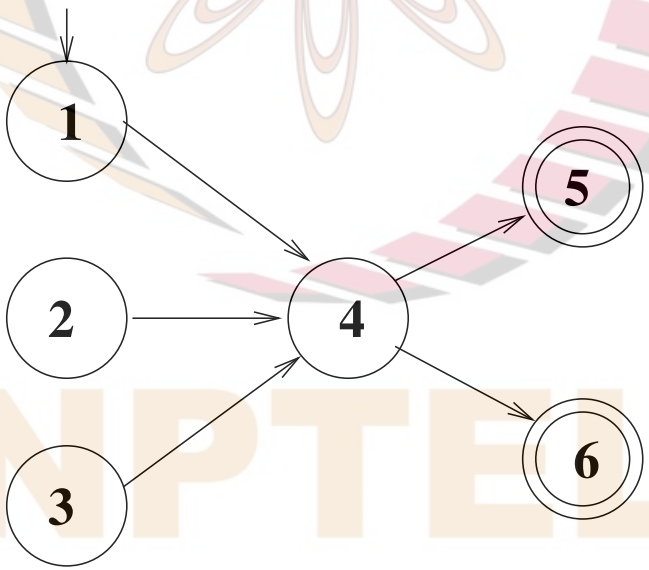


- Node coverage: $TR = \{1, 2, 3\}$, test path $= [1, 2, 3]$.
- Edge coverage: $TR = \{(1, 2), (1, 3), (2, 3)\}$, test paths $= \{[1, 2, 3], [1, 3]\}$.

Note: Node coverage and edge coverage are different only when there is an edge in another sub-path between a pair of nodes, like the above graph.

Covering Multiple Edges: Edge-Pair Coverage

- Edge-Pair Coverage (EPC): TR contains each reachable path of length up to 2, inclusive, in G.
- Paths of length up to 2 correspond to pairs of edges.
- Again, the phrase "length up to 2" ensures edge-pair coverage holds for graphs with less than 2 edges.



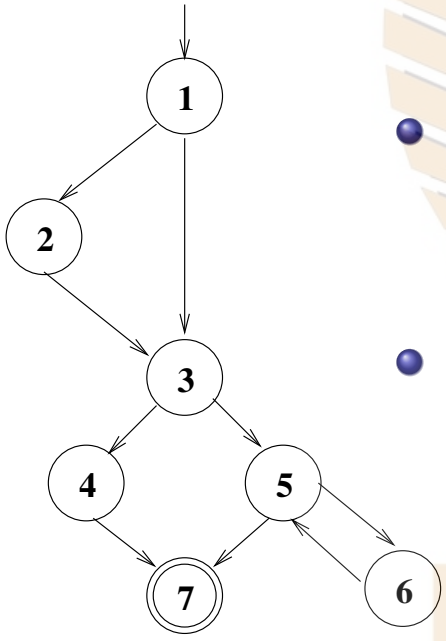
- $TR = \{[1, 4, 5], [2, 4, 5], [3, 4, 5], [1, 4, 6], [2, 4, 6], [3, 4, 6]\}$.
- Test paths are the same as above.

Covering Multiple Edges, contd.

- An extension of edge-pair coverage is to consider all paths.
- Complete path coverage: TR contains all paths in G.
- Unfortunately, this is can be an infeasible test requirement. Also, it may not be a useful test requirement.
- Specified path coverage: TR contains a set S of paths, where S is specified by the user/tester.

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Coverage criteria seen so far: Another example



- Node coverage: $TR = \{1, 2, 3, 4, 5, 6, 7\}$, Test paths: $\{[1, 2, 3, 4, 7], [1, 3, 5, 6, 5, 7]\}$.
- Edge coverage: $TR = \{(1, 2), (1, 3), (2, 3), (3, 4), (4, 7), (3, 5), (5, 6), (6, 5), (5, 7)\}$ Test paths: $\{[1, 2, 3, 4, 7], [1, 3, 5, 6, 5, 7]\}$.
- Edge-pair coverage: $TR = \{[1, 2, 3], [1, 3, 4], [1, 3, 5], [2, 3, 4], [2, 3, 5], [3, 4, 7], \ldots\}$, Test paths: $\{[1, 2, 3, 4, 7], [1, 2, 3, 5, 7], [1, 3, 4, 7], [1, 3, 5, 6, 5, 6, 5, 7]\}$.
- Complete path coverage: $TR = \{[1, 2, 3, 4, 7], [1, 2, 3, 5, 7], [1, 2, 3, 5, 6, 5, 7], \ldots\}$.

Complete and Specified Path Coverage

- If a graph contains a loop, it has an *infinite* number of paths and hence complete path coverage is infeasible.
- What will be a good notion of specified path coverage in the presence of loops?
- Loops have boundary conditions and repeated executions. Effective test cases will not execute the loop for every iteration.
- Ideally, we need to have test cases that *cover* the loop:
  - Execute the loop at its boundary conditions— skip the loop execution.
  - Execute the loop *once* for normal iterations.
- The notion of prime paths came into existance for working with loops.

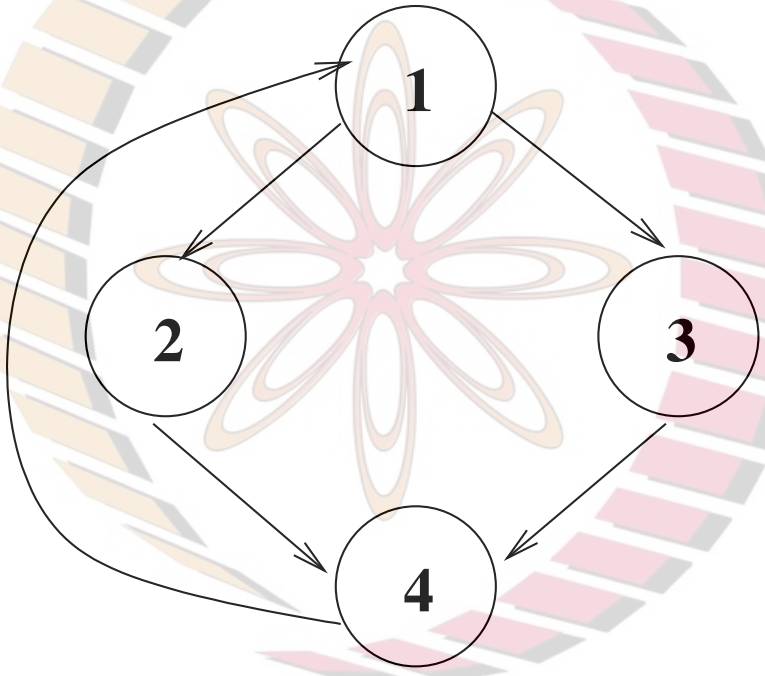
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Prime Paths in Graphs

- A path from node n_i to n_j is simple if no node appears more than once, except possibly the first and last node.
  - No internal loops.
  - A loop is a simple path.
- A prime path is a simple path that does not appear as a proper subpath of any other simple path.

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Simple paths and prime paths: Example



- Simple paths: [1,2,4,1], [1,3,4,1], [2,4,1,2], [2,4,1,3], [3,4,1,2], [3,4,1,3], [4,1,2,4], [4,1,3,4], [1,2,4], [1,3,4], [2,4,1], [3,4,1], [4,1,2], [4,1,3], [1,2], [1,3], [2,4], [3,4], [4,1], [1], [2], [3], [4]
- Prime paths: [2,4,1,2], [2,4,1,3], [1,3,4,1], [1,2,4,1], [3,4,1,2], [4,1,3,4], [4,1,2,4], [3,4,1,3]

Prime path coverage

- Prime path coverage: TR contains each prime path in G.
- Ensures that loops are skipped as well as executed.
- By touring all paths of length 0 and 1, it subsumes node and edge coverage.
- May or may not subsume edge-pair coverage.

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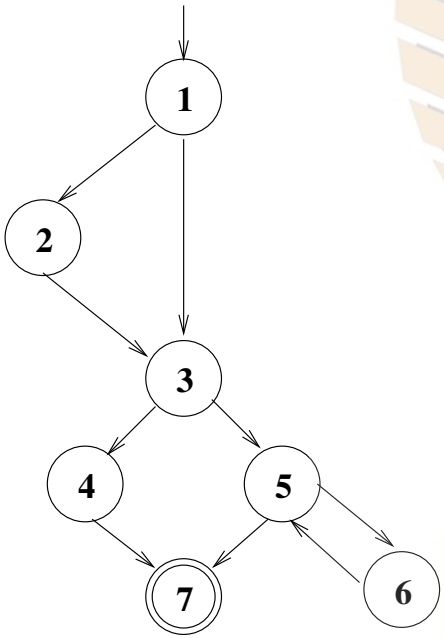
Prime path coverage vs. edge-pair coverage



- In graphs where there are self loops, edge-pair coverage requires the self loop to be visited.
- For e.g., in the above graph, *TR* for edge-pair coverage will be $\{[1,2,3],[1,2,2],[2,2,3],[2,2,2]\}$.
- Some of these are prime paths/simple paths.
- *TR* for prime path coverage for the above example is $\{1,2,3],[2,2]\}$.

Prime path coverage and loops in graphs

Prime paths capture the notion of *covering* a loop well.



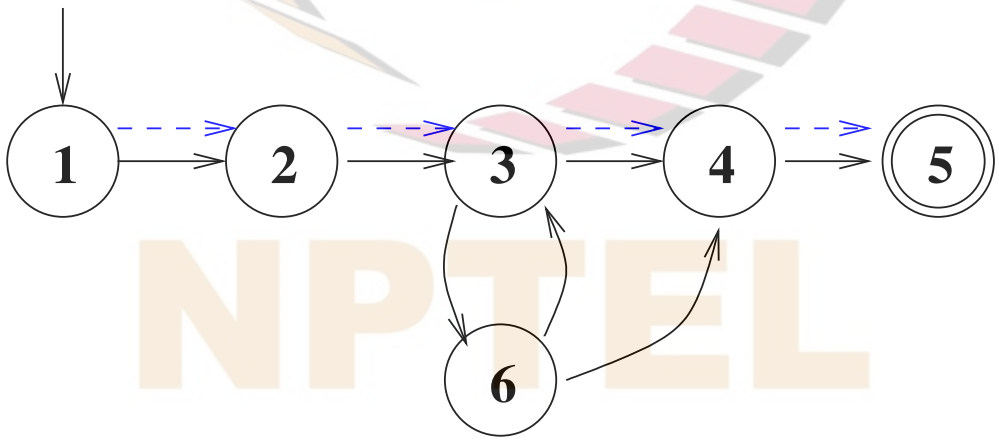
- There are nine prime paths.
- They correspond to

1,3,5,7 : Skipping the loop,
1,3,5,6 : Executing the loop once, and
6,5,6 : Executing the loop more than once.

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Implementing test requirements for prime path coverage

- Prime paths, by definition, do not have internal loops.
- In many cases, it might be impossible to meet the test requirement of prime path coverage without internal loops.
- That is, test paths that meet prime path coverage TR need to have internal loops to make prime path coverage feasible.



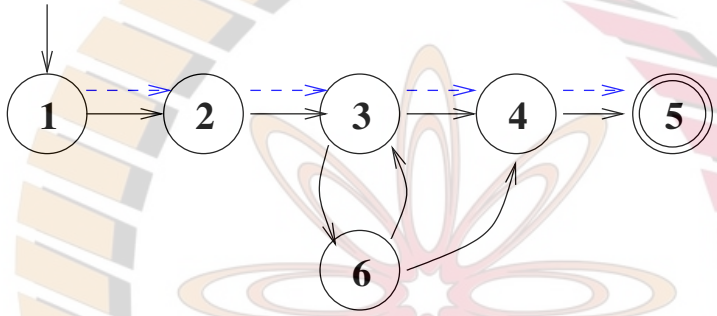
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Touring, side trips and detours

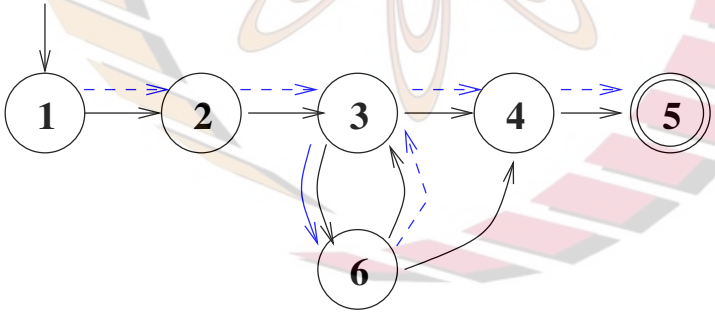
- Tour: A test path p tours sub-path q if q is a sub-path of p.
- Tour with side trips: A test path p tours sub-path q with sidetrips iff every edge in q is also in p in the same order.
  - The tour can include a sidetrip, as long as it comes back to the same node.
- Tour with detours: A test path p tours sub-path q with detours iff every node in q is also in p in the same order.
  - The tour can include a detour from node n, as long as it comes back to the prime path at a successor of n.

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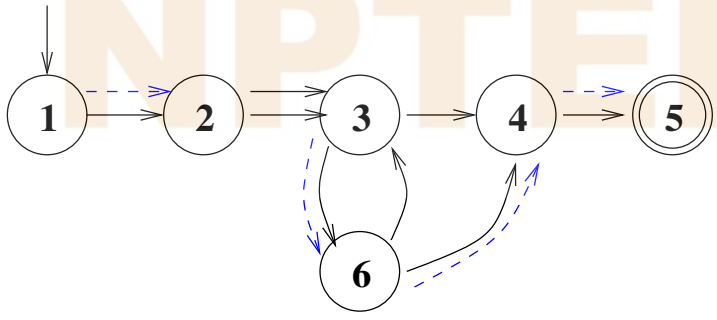
Touring, side trips and detours: Example



Touring the prime path without side trips and detours



Touring the prime path with a side trip



Touring the prime path with a detour

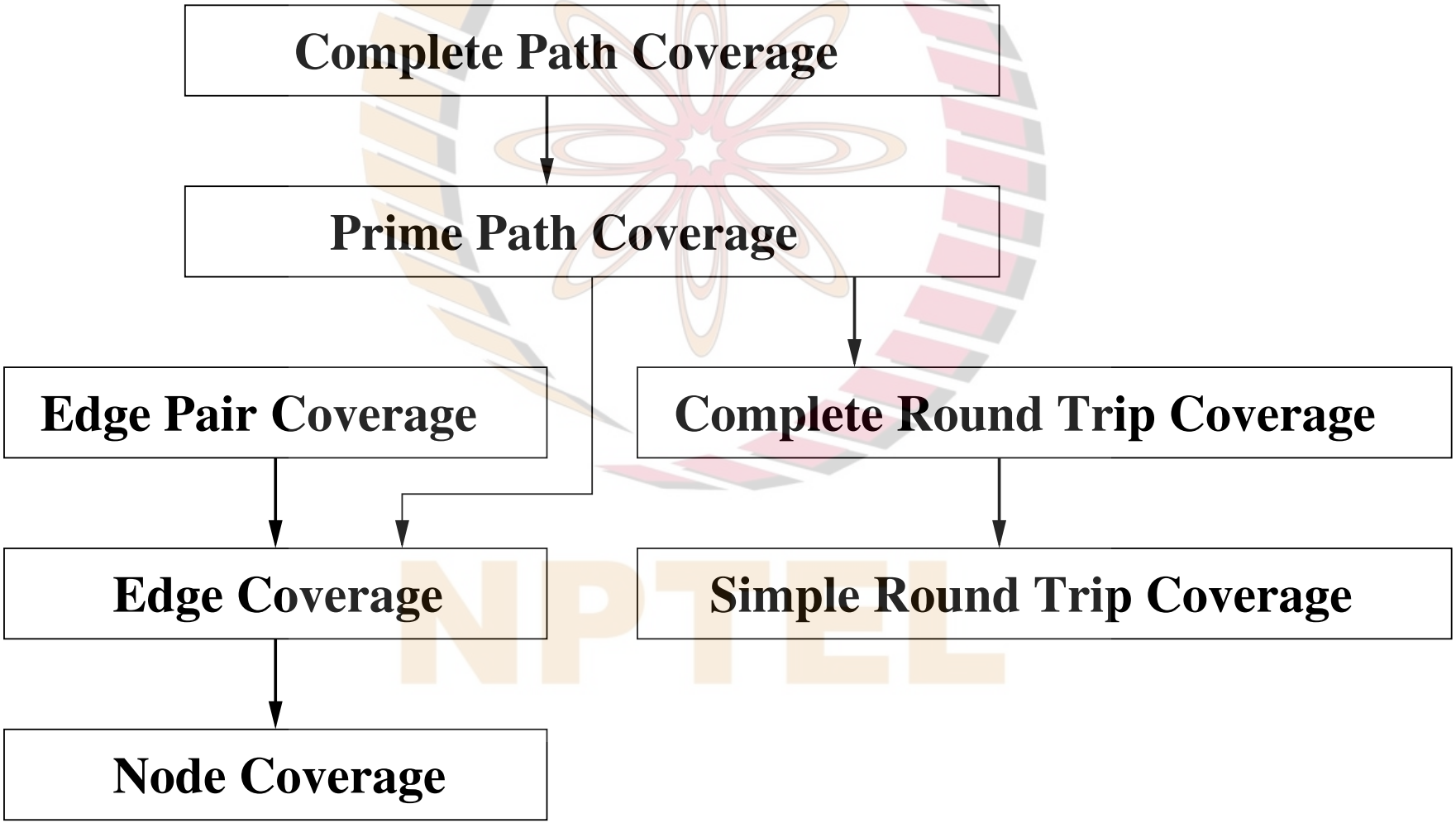
Infeasible test requirements

- Some test requirements related to graph structural coverage can be infeasible.
- It is undecidable to check if many structural coverage requirements are feasible or not.
- Typically, when side trips are not allowed, many structural coverage criteria have infeasible test requirements.
- However, always allowing side trips weakens the test criteria.
- Best Effort Touring: Satisfy as many test requirements as possible without sidetrips. Allow sidetrips to try to satisfy remaining test requirements.

Round trips

- Round trip path: A prime path that starts and ends at the same node.
- Simple round trip coverage: TR contains at least one round trip path for each reachable node in G that begins and ends in a round trip path.
- Complete round trip coverage: TR contains all round trip paths for each reachable node in G.
- The above two criteria omit nodes and edges that are not in round trips.
- Hence, they do not subsume edge-pair, edge or node coverage.

Structural Coverage Criteria Subsumption



Credits

Part of the material used in these slides are derived from the presentations of the book Introduction to Software Testing, by Paul Ammann and Jeff Offutt.

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